



IALA S-201 Product Specification

Edition 2.0.0 – May 2025

IALA AtoN Product Specification

Overview

S-201 Aids to Navigation Information Product Specification - Overview

1 Introduction

The Aids to Navigation (AtoN) Information Product Specification provides a common structure for the exchange of information about AtoNs. This includes buoys, beacons, racons, lights, sound signals and AIS. The product contains the positions, properties, operational status and general comments related to an AtoN.

The Product Specification can be used to exchange AtoN information in a consistent form between Lighthouse Authorities, Hydrographic Offices and other organizations, including commercial and professional agencies.

2 Contents of the Specification

Main document

The AtoN Information Product Specification follows the format set out in the IHO Standard S-100 for geo-spatial information. This describes the scope, data content and structure, specifies procedures for data maintenance and quality and details the encoding of the data.

3 Annexes

The annexes provide information about use of the specification, as described in the following table.

Title	Description
Annex A DCEG	The Data Classification and Encoding Guide has been developed to produce consistent and standardized instructions for encoding S-201 datasets.
Annex B Data Product Format, Schemas	This package contains the S-100 Part 10b compliant GML schemas for the specification of the data product encoding for S-201 datasets.
Annex C1 Feature Catalogue	This package is generated from the accompanying XML S-201 feature catalogue and presents the feature catalogue in a reader-friendly form. In its current form, it is intended to facilitate reviews of the content of the XML feature catalogue.
Annex C2 Feature Catalogue XML	The package contains the S-201 feature catalogue in a machine-readable XML format, which complies with the IHO S-100 feature catalogue schema. This document is meant for use by systems that are intended to read/write S-201 datasets.
Annex D Portrayal Catalogue	The package contains the S-201 portrayal catalogue.

4 Method of Use

Given the complexity of the specification it is envisaged that some form of user interface will be needed. The developers of this interface will need to understand the full scope and content of the

specification and will be required to validate it, to ensure that the data output is identical to the data input. However, the interface operator should not need to understand the specification beyond its purpose. They should be given on-screen prompts for data inputs and the output format would be similarly annotated to make clear the meaning of the resulting data.